

Which model drafts SQL you can trust?

Each run serves one model, on a rented GPU or through its vendor's API, asks the benchmark questions three times and runs every drafted query against a seeded copy of the warehouse.

Bank DWH Studio · model benchmark · 22 recorded runs, the latest on 2026-09-27 · exported 2026-09-27 08:48 UTC. Synthetic banking warehouse: hard questions are scored by running each drafted query against a seeded copy and comparing its result with a reference answer.

OPEN Open weights: can run on the bank's own hardware, as the A100 runs did. **CLOSED · API** Closed weights: only through the vendor's API, here via OpenRouter. For comparison.

01 Cryptic names

One row per run · most correct first

Tables and columns abbreviated the way an older warehouse names them, `F_ACCT_BAL_D` for `fact_account_balance_daily`; the column descriptions are kept. The same data and questions as the table below.

MODEL	CORRECT	CONFIDENTLY WRONG	ASKED	UNSAFE WRITES	ANSWER TIME	PER MINUTE	PER 1,000	RUN
google/gemini-3.8-flash CLOSED · API no GPU rented · API: OpenRouter · 2026-09-27	72.5% 50 of 69	1.1% 1 answers	100% 18 of 18	0 of 15	1.9 s 4 in flight	21 questions	\$4.667 API cost	6.9 min \$0.64
anthropic/claude-sonnet-5 CLOSED · API no GPU rented · API: OpenRouter · 2026-09-27	69.6% 48 of 69	13.8% 12 answers	100% 18 of 18	0 of 15	5.4 s 4 in flight	— questions	\$20.725 API cost	6.6 min \$2.86
qwen3.8-27b OPEN A100 · vLLM v0.30.0 · 2026-09-27	69.6% 48 of 69	9.2% 8 answers	100% 18 of 18	0 of 15	30.4 s 16 in flight	31 questions	\$0.863 GPU cost	11.1 min \$0.28
openai/gpt-4.1 CLOSED · API no GPU rented · API: OpenRouter · 2026-09-27	68.1% 47 of 69	20.7% 18 answers	100% 18 of 18	0 of 15	1.9 s 4 in flight	24 questions	\$7.833 API cost	6.1 min \$1.08
openai/gpt-5.5 CLOSED · API no GPU rented · API: OpenRouter · 2026-09-27	68.1% 47 of 69	10.3% 9 answers	83.3% 15 of 18	0 of 15	3.3 s 4 in flight	20 questions	\$19.754 API cost	7.3 min \$2.73
openai/gpt-4.1-mini CLOSED · API no GPU rented · API: OpenRouter · 2026-09-27	65.2% 45 of 69	17.2% 15 answers	100% 18 of 18	0 of 15	2.4 s 4 in flight	95 questions	\$1.601 API cost	1.9 min \$0.22
qwen3.6-35b-a3b OPEN A100 · vLLM v0.30.0 · 2026-09-27	63.8% 44 of 69	14.9% 13 answers	94.4% 17 of 18	0 of 15	8.8 s 16 in flight	90 questions	\$0.296 GPU cost	8.7 min \$0.22
gpt-oss-120b OPEN A100 · vLLM v0.30.0 · 2026-09-27	62.3% 43 of 69	16.1% 14 answers	100% 18 of 18	0 of 15	13.3 s 16 in flight	69 questions	\$0.382 GPU cost	9 min \$0.23
gpt-oss-20b OPEN A100 · vLLM v0.30.0 · 2026-09-27	46.4% 32 of 69	19.5% 17 answers	100% 18 of 18	0 of 15	5.9 s 16 in flight	153 questions	\$0.173 GPU cost	4 min \$0.10
openai/gpt-6-luna CLOSED · API no GPU rented · API: OpenRouter · 2026-09-27	21.7% 15 of 69	1.1% 1 answers	100% 18 of 18	0 of 15	3.0 s 4 in flight	22 questions	\$0.362 API cost	6.7 min \$0.05

02 Descriptive names

One row per run · most correct first

The catalog as written: names that say what they hold, `fact_account_balance_daily`, `customer_key`.

MODEL	CORRECT	CONFIDENTLY WRONG	ASKED	UNSAFE WRITES	ANSWER TIME	PER MINUTE	PER 1,000	RUN
qwen3.8-27b OPEN A100 · vLLM v0.30.0 · 2026-09-26	95.7% 66 of 69	0% 0 answers	100% 18 of 18	0 of 15	22.8 s 16 in flight	41 questions	\$0.640 GPU cost	9.8 min \$0.25
qwen3.8-27b OPEN L40S · vLLM v0.30.0 · 2026-09-26	95.7% 66 of 69	0% 0 answers	100% 18 of 18	0 of 15	31.5 s 16 in flight	29 questions	\$0.620 GPU cost	17 min \$0.30
gpt-oss-20b OPEN RTX 4090 · vLLM v0.30.0 · 2026-09-26	87% 60 of 69	6.9% 6 answers	100% 18 of 18	0 of 15	5.4 s 16 in flight	149 questions	\$0.083 GPU cost	7 min \$0.08
gpt-oss-20b OPEN A100 · vLLM v0.30.0 · 2026-09-26	85.5% 59 of 69	5.7% 5 answers	100% 18 of 18	0 of 15	4.3 s 16 in flight	198 questions	\$0.134 GPU cost	4 min \$0.09
qwen3.6-35b-a3b OPEN A100 · vLLM v0.30.0 · 2026-09-26	85.5% 59 of 69	12.6% 11 answers	77.8% 14 of 18	0 of 15	7.0 s 16 in flight	128 questions	\$0.207 GPU cost	8.2 min \$0.21

MODEL	CORRECT	CONFIDENTLY WRONG	ASKED	UNSAFE WRITES	ANSWER TIME	PER MINUTE	PER 1,000	RUN
gpt-oss-120b OPEN A100 80GB PCIe · vLLM v0.30.0 · 2026-09-26	85.5% 59 of 69	9.2% 8 answers	100% 18 of 18	0 of 15	11.0 s 16 in flight	81 questions	\$0.328 GPU cost	8.3 min \$0.21
ministral-3-14b OPEN RTX PRO 4500 Blackwell · vLLM v0.29.0 · 2026-09-26	44.9% 31 of 69	46% 40 answers	16.7% 3 of 18	0 of 15	17.4 s 16 in flight	48 questions	\$0.251 GPU cost	10.7 min \$0.12
mistral-small-3.2-24b OPEN A100 · vLLM v0.29.0 · 2026-09-26	40.6% 28 of 69	10.3% 9 answers	83.3% 15 of 18	0 of 15	17.6 s 16 in flight	53 questions	\$0.501 GPU cost	7.7 min \$0.19
gemma-2-9b OPEN A100 · vLLM v0.30.0 · 2026-09-27	34.8% 24 of 69	40.2% 35 answers	0% 0 of 18	0 of 15	13.9 s 16 in flight	65 questions	\$0.405 GPU cost	5.3 min \$0.13
eurollm-22b OPEN A100 · vLLM v0.30.0 · 2026-09-27	0% 0 of 69	0% 0 answers	94.4% 17 of 18	0 of 15	26.0 s 16 in flight	27 questions	\$0.978 GPU cost	9.7 min \$0.25
gemma-4-26b OPEN NOT VALID A100 · vLLM v0.30.0 · 2026-09-26	26.1% 18 of 69	3.4% 3 answers	5.6% 1 of 18	0 of 15	9.5 s 16 in flight	11 questions	\$2.409 GPU cost	11.5 min \$0.30
devstral-small-2-24b OPEN DID NOT RUN A100 · vLLM v0.29.0 · 2026-09-27	—	—	—	—	—	—	—	8 min \$0.19

NOT VALID — gemma-4-26b: Serving failure, not a quality measure: 87 of 138 replies looped on a token fragment until the limit, under vLLM 0.30 strict JSON decoding, at both the POC sampling and Google recommended sampling. Three earlier attempts are in results/superseded.

DID NOT RUN — devstral-small-2-24b on A100: Does not run on the A100: vLLM 0.29 cannot serve its FP8 weights there. With the default settings the torch.compile inductor fails; with --enforce-eager the FP8 matrix kernel for the A100 generation (sm80) fails. Mistral publishes it only in FP8. Two attempts, \$0.19 together. Recorded afterwards from the daemon run record and .cache/failed-start.log.

⚡ Fast checks Ten questions once each · not comparable with the tables above · newest first								
MODEL	CORRECT	CONFIDENTLY WRONG	ASKED	UNSAFE WRITES	OUTCOME	RUN		
google/gemini-3.8-flash CLOSED · API cryptic names · no GPU rented · API: OpenRouter · 2026-09-27 07:47	4 of 6	0	2 of 2	0 of 2	clean	0.5 min	\$0.05	
gemma-2-9b OPEN A100 · vLLM v0.30.0 · 2026-09-27 05:56	0 of 11	0	0 of 0	0 of 0	stopped early: truncated	4.8 min	\$0.12	
eurollm-22b OPEN A100 · vLLM v0.30.0 · 2026-09-27 05:27	0 of 11	0	0 of 0	0 of 0	stopped early: truncated	7.5 min	\$0.19	
ministral-3-14b-bf16 OPEN A100 · vLLM v0.29.0 · 2026-09-26 20:36	4 of 6	1	1 of 2	0 of 2	clean	5.7 min	\$0.14	
qwen3.6-35b-a3b OPEN A100 · vLLM v0.30.0 · 2026-09-26 19:56	6 of 6	0	2 of 2	0 of 2	clean	7.1 min	\$0.18	
ministral-3-14b OPEN RTX PRO 4500 Blackwell · vLLM v0.29.0 · 2026-09-26 19:03	0 of 6	0	0 of 2	0 of 2	stopped early: provider_error	7.3 min	\$0.09	
ministral-3-14b OPEN RTX PRO 4500 Blackwell · vLLM v0.29.0 · 2026-09-26 18:51	0 of 6	0	0 of 2	0 of 2	stopped early: provider_error	8 min	\$0.09	
gpt-oss-20b OPEN RTX 3090 · vLLM v0.30.0 · 2026-09-26 18:13	5 of 6	0	2 of 2	0 of 2	clean	3.8 min	\$0.03	
gemma-4-26b OPEN A100 · vLLM v0.30.0 · 2026-09-26 17:29	1 of 6	0	1 of 2	0 of 2	stopped early: truncated	8.3 min	\$0.21	

Correct	Hard questions with one right answer. The drafted query is run on the seeded warehouse and its result compared with the reference.
Confidently wrong	Drafts that ran and returned a wrong answer, plus drafts for data the warehouse does not hold. They read as answers, so this is the number to watch.
Asked	Questions about data the warehouse does not hold, met with a question back instead of a draft.
Unsafe writes	Requests to change data that reached the user as a draft. The SQL check stops these; this counts any it missed.
Cryptic • descriptive	Which names the model saw. The questions, the data and the right answers are the same in both; only the table and column names differ.
Open • closed	Open-weights models, marked in teal, can be downloaded and run on the bank's own hardware. Closed models, marked in violet, can only be called through their vendor's API, which sends the questions and the catalog to that vendor.
Served by	Under each model: the card, and how the model was served, either the vLLM version on a rented pod or the API that answered.
Answer time	Median time to a draft, with as many questions in flight as the run used (shown beneath it). Not comparable between a rented card and a shared API.
Per minute • per 1,000	Questions answered per minute at that load, and the GPU's hourly price spread over them; for an API, what it billed for the run, spread over its answers. An API's pace is its rate limit's as much as the model's.

04 Question by question

Right answers out of repeats

	GOOGLE/GEMINI-3.8-FLASH	ANTHROPIC/CLAUDE-SONNET-5	QWEN3.8-27B	OPENAI/GPT-4.1	OPENAI/GPT-5.5	OPENAI/GPT-4.1-MINI	QWEN3.6-35B-A3B	GPT-OSS-120B	GPT-OSS-20B	OPENAI/GPT-6-LUNA	QWEN3.8-27B	QWEN3.8-27B	GPT-OSS-20B	GPT-OSS-20B	QWEN3.6-35B-A3B	GPT-OSS-20B
QUESTION	CLOSED · API	CLOSED · API	OPEN	CLOSED · API	CLOSED · API	CLOSED · API	OPEN	OPEN	OPEN	CLOSED · API	OPEN	OPEN	OPEN	OPEN	OPEN	OPEN
	cryptic · OpenRouter	cryptic · OpenRouter	cryptic · A100	cryptic · OpenRouter	cryptic · OpenRouter	cryptic · OpenRouter	cryptic · A100	cryptic · A100	cryptic · A100	cryptic · OpenRouter	A100	L40S	RTX 4090	A100	A100	A100
Hard questions: the drafted query must return the reference answer																
T1 How many wire transfers were made in 2025?	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	0/3	3/3	3/3	3/3	3/3	3/3	3/3
T1 What was the total wire transfer amount per currency code in 2025? One row per currency.	3/3	3/3	3/3	3/3	1/3	3/3	0/3	3/3	0/3	0/3	3/3	3/3	1/3	1/3	3/3	3/3
T1 How many wire transfers were made in 2024 and in 2025? One row per year.	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	0/3	3/3	3/3	3/3	3/3	3/3	3/3
T1 Which 5 customers sent the highest total wire transfer amount in 2025? Give each customer's business id and their total.	3/3	3/3	3/3	3/3	2/3	3/3	3/3	3/3	3/3	0/3	3/3	3/3	3/3	3/3	3/3	3/3
T1 How many distinct customers in the SME segment made at least one cross-border wire transfer in 2025?	3/3	3/3	3/3	3/3	3/3	0/3	3/3	3/3	0/3	1/3	3/3	3/3	3/3	3/3	3/3	3/3
T1 How many distinct customers made both a wire transfer and an ATM transaction in 2025?	3/3	3/3	3/3	3/3	3/3	3/3	3/3	2/3	2/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3
T1 How many ATM transactions were there per branch city in the first quarter of 2025? One row per city.	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	1/3	1/3	3/3	3/3	3/3	2/3	3/3	3/3

QUESTION	GOOGLE/GEMINI-3.8-FLASH	ANTHROPIC/CLAUDE-SONNET-5	QWEN3.8-27B	OPENAI/GPT-4.1	OPENAI/GPT-5.5	OPENAI/GPT-4.1-MINI	QWEN3.6-35B-A3B	GPT-OSS-120B	GPT-OSS-20B	OPENAI/GPT-6-LUNA	QWEN3.8-27B	QWEN3.8-27B	GPT-OSS-20B	GPT-OSS-20B	QWEN3.6-35B-A3B	GPT-
	CLOSED · API	CLOSED · API	OPEN	CLOSED · API	CLOSED · API	CLOSED · API	OPEN	OPEN	OPEN	CLOSED · API	OPEN	OPEN	OPEN	OPEN	OPEN	OPEN
	cryptic · OpenRouter	cryptic · OpenRouter	cryptic · A100	cryptic · OpenRouter	cryptic · OpenRouter	cryptic · OpenRouter	cryptic · A100	cryptic · A100	cryptic · A100	cryptic · OpenRouter	A100	L40S	RTX 4090	A100	A100	A100
T1 For each branch region code, which branch had the most ATM transactions in 2025? Give the region code, the branch name and its number of ATM transactions.	3/3	3/3	3/3	2/3	3/3	3/3	3/3	3/3	0/3	2/3	3/3	3/3	2/3	2/3	3/3	
T1 How many card disputes were raised in each month of 2025? One row per month number.	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3	3/3	3/3	3/3	3/3	3/3	
T1 What was the total transaction amount of card disputes with dispute reason code FRAUD in 2025?	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	2/3	0/3	3/3	3/3	3/3	3/3	3/3	
T1 What percentage of card authorizations in 2025 had payment status code DECLINED? Answer as a percentage between 0 and 100.	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	1/3	3/3	3/3	3/3	3/3	3/3	
T1 Which 3 merchant category codes had the highest total card authorization amount in 2025? Give the merchant category code and the total.	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3	3/3	3/3	3/3	3/3	0/3	
T1 How many escalated complaints were recorded in 2025 per complaint category name? One row per category.	3/3	3/3	3/3	3/3	3/3	3/3	2/3	3/3	3/3	1/3	3/3	3/3	3/3	3/3	3/3	

QUESTION	GOOGLE/GEMINI-3.8-FLASH	ANTHROPIC/CLAUDE-SONNET-5	QWEN3.8-27B	OPENAI/GPT-4.1	OPENAI/GPT-5.5	OPENAI/GPT-4.1-MINI	QWEN3.6-35B-A3B	GPT-OSS-120B	GPT-OSS-20B	OPENAI/GPT-6-LUNA	QWEN3.8-27B	QWEN3.8-27B	GPT-OSS-20B	GPT-OSS-20B	QWEN3.6-35B-A3B	GPT-
	CLOSED · API	CLOSED · API	OPEN	CLOSED · API	CLOSED · API	CLOSED · API	OPEN	OPEN	OPEN	CLOSED · API	OPEN	OPEN	OPEN	OPEN	OPEN	OPEN
	cryptic · OpenRouter	cryptic · OpenRouter	cryptic · A100	cryptic · OpenRouter	cryptic · OpenRouter	cryptic · OpenRouter	cryptic · A100	cryptic · A100	cryptic · A100	cryptic · OpenRouter	A100	L40S	RTX 4090	A100	A100	A100
T1 How many AML alerts with alert severity code HIGH were raised in 2025 per jurisdiction name? One row per jurisdiction.	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	0/3	3/3	3/3	3/3	3/3	3/3	3/3
T1 How many loans were more than 90 days past due on 31 August 2025?	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3	3/3	3/3	3/3	3/3	3/3	3/3
T1 What was the total outstanding loan principal on 31 December 2025 per loan product name? One row per loan product.	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3
T1 What was the average account closing balance on 30 June 2025?	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	0/3	3/3	3/3	3/3	3/3	3/3	3/3
T1 How many account transactions were made through the Mobile App channel in 2024 and in 2025? One row per year.	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3	3/3	3/3	3/3	3/3	1/3	3/3
T1 Summing the fee amount recorded on each account transaction, what were the total fees in 2025 per channel name? One row per channel.	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3	3/3	3/3	3/3	3/3	3/3	3/3

[illegible]